



Vivid Color-Mix Tool

- Quick Start Guide -

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CATALOGUE

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**Vivid
Color-Mix
Tool**



INTRODUCTION

- Vivid Color-Mix Tool provides users a realistic way to mix colors.
- Here, yellow mix with blue is green, not gray in traditional sRGB.
- It's realtime, lightweight and easy to use.
- Try Vivid Color-Mix Tool now!

**Vivid
Color-Mix
Tool**

sRGB Color Mix

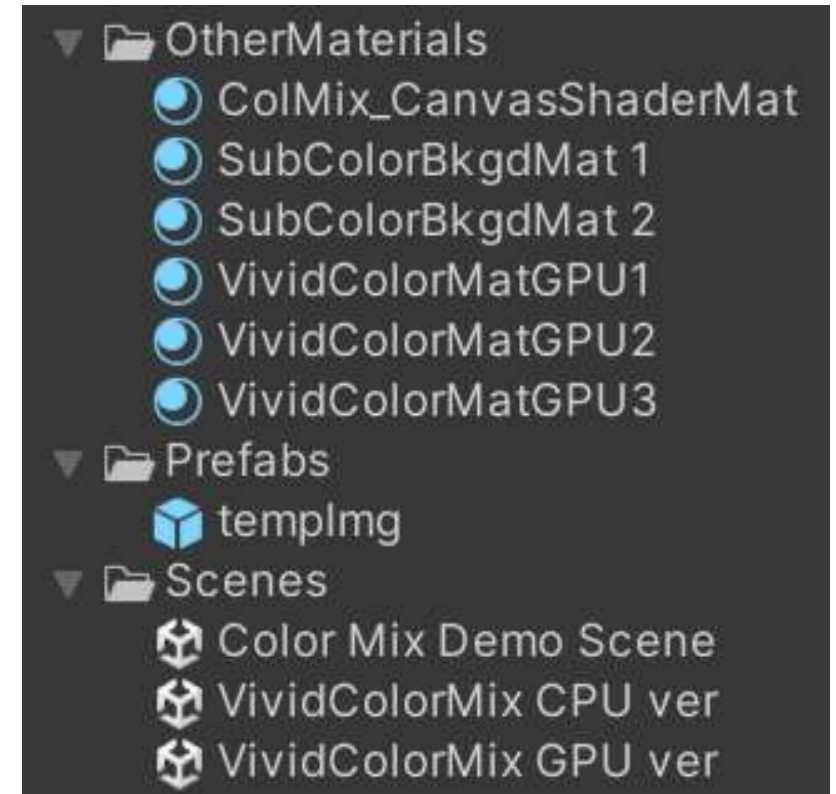


Vivid Color Mix Tool



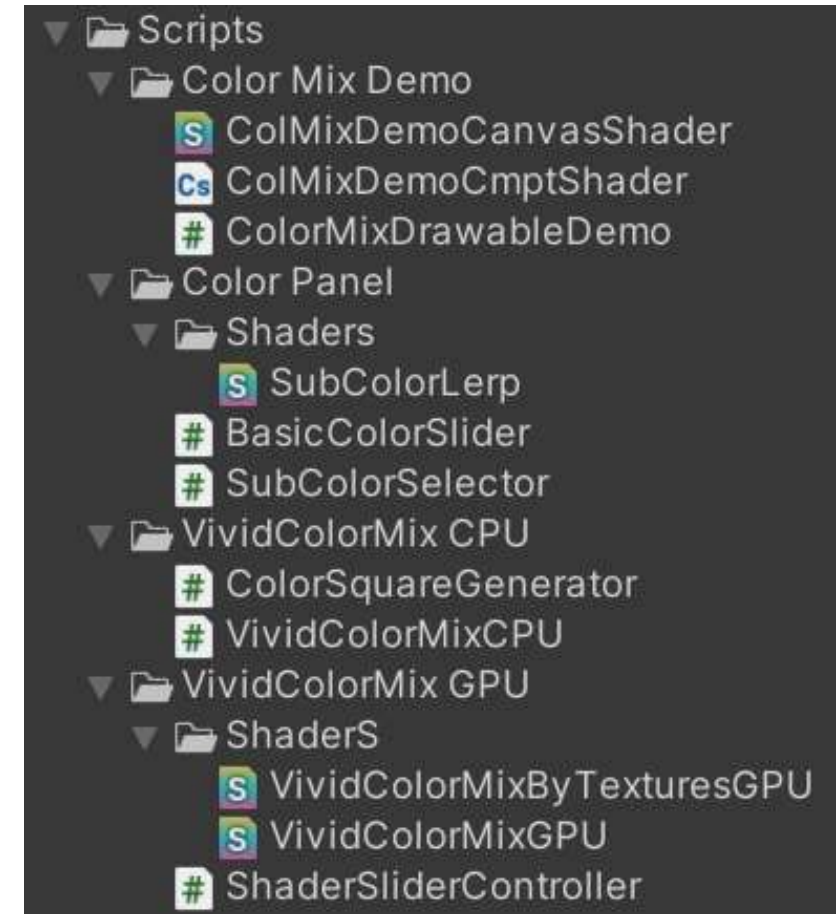
PROJECT FILES

- In the **OtherMaterials** folder, some **materials** are used for a color picker, shader script and paint demo of Vivid Color-Mix Tool.
- In the **Prefabs** folder you can see a templmg **prefab**, it is used for generating the color bands.
- In the **Scenes** folder, there are 3 **use cases** for you to try, they are drawable demo and color mix test realized in both C# script and unity shader.



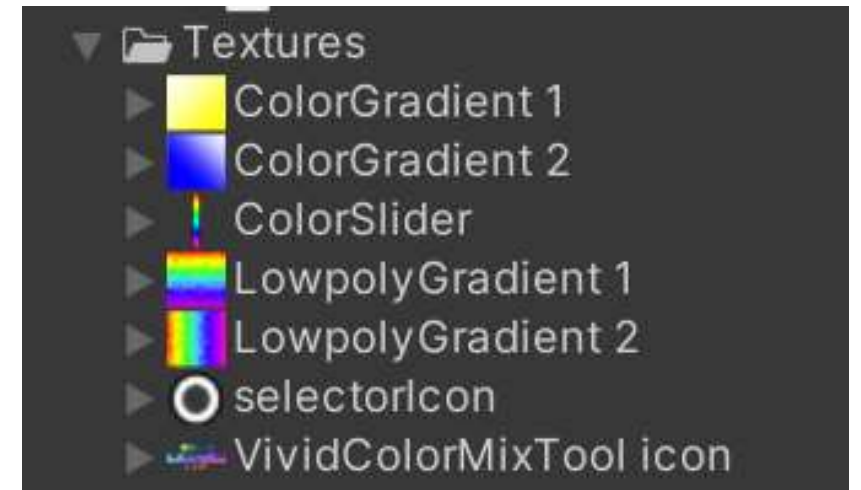
PROJECT FILES

- In the **Scripts** folder:
- **ColorMixDemo** folder contains scripts, shader and compute shader used in the drawable demo, you can find them in **Scene ColorMixDemoScene**.
- **ColorPanel** folder contains scripts & shader used in the color picker part, you can find them in **Scene VividColorMixCPUver**.
- **VividColorMixCPU** folder contains the C# version of the Vivid Color-Mix algorithm, and script used to generate the color bands.
- **VividColorMixGPU** folder contains the **shader version** of the Vivid Color-Mix algorithm (pure color mix and texture color mix), and a slider controller script, all of them are used in **Scene VividColorMixGPUver**.



PROJECT FILES

- In the **Textures** folder, there are textures used in this package's scenes.
- Some of them are **background images or icons** of some game object components.
- And others are **textures** used for the **shader version** (more exactly, the texture color mix version) of Vivid Color-Mix Tool, you can find them in **Scene VividColorMixGPUver**.



USE CASE (C# SCRIPT)

- In the Scene VividColorMixCPUver, you can find 2 color pickers and 2 color bands.
- The 2 color bands respectively represent color mix in RGB mode & VividColorMix mode.
- You can change color pickers' color and try other colors' mix result.



USE CASE (C# SCRIPT)

- The source code of the main Vivid Color-Mix algorithm is also provided.
- It is defined in the script named VividColorMixCPU, and called in script ColorSquareGenerator.
- The input value are: 2 colors that will be mixed, and an interpolation parameter T.

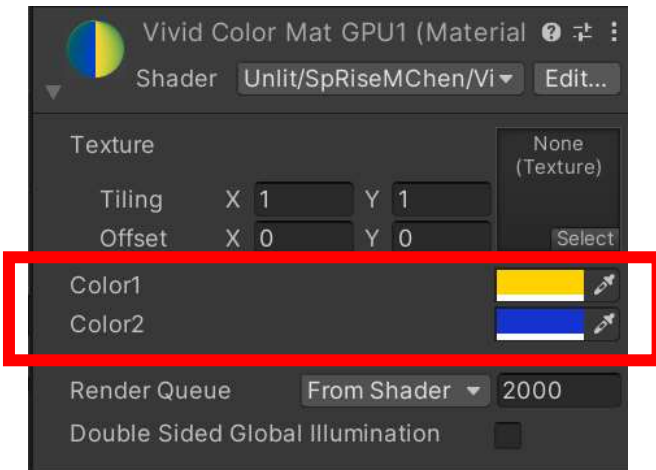
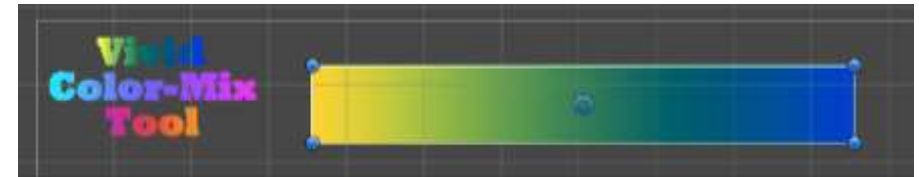
```
Color col1 = img1.color, col2 = img2.color;

switch (mixType)
{
    case MixType.sRGBMix:
        for (int i = 0; i <= steps; ++i)
        {
            float t = (float)i / steps;
            imageList[i].color = Color.Lerp(col1, col2, t);
        }
        break;
    case MixType.vividColorMix:
        for (int i = 0; i <= steps; ++i)
        {
            float t = (float)i / steps;
            imageList[i].color = VividColorMixCPU.Vivid_Color_Mix(col1, col2, t);
        }
        break;
    default:
        break;
}
```



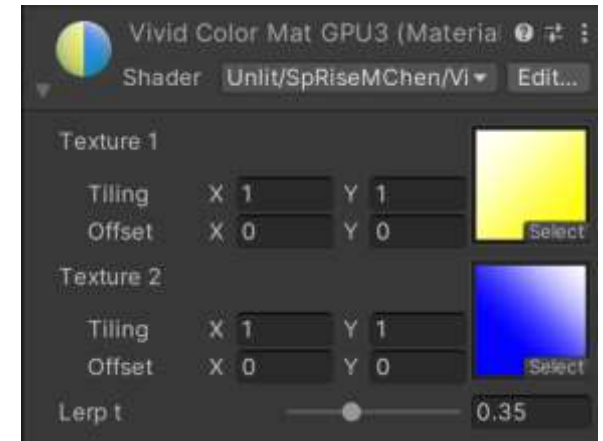
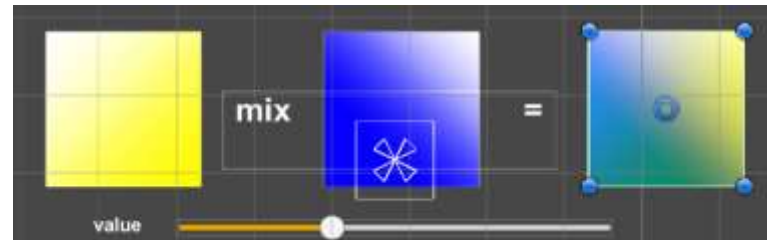
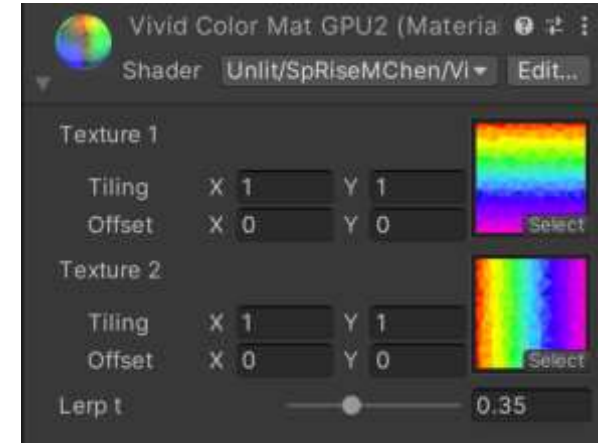
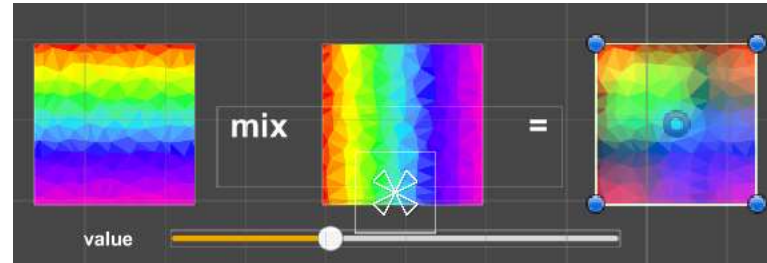
USE CASE (UNITY SHADER SCRIPT)

- In the Scene **VividColorMixGPUver**, you can find 1 color band and 2 texture mix equations.
- Let's talk about the color band first, you can **click the color band game object in Scene Window**, and check out its **material/shader properties**.
- Then, you can change the color properties in the shader to change the mix result.



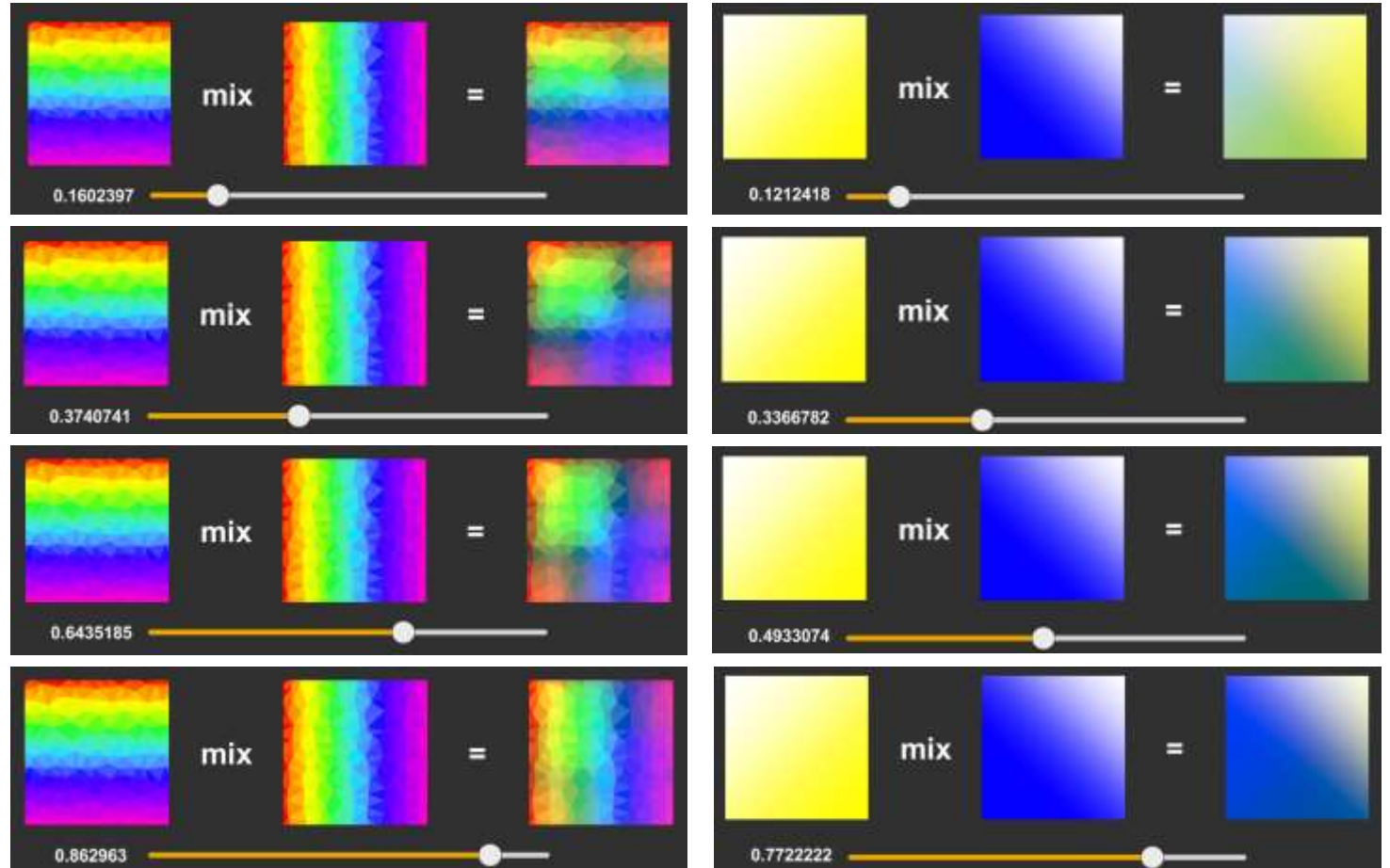
USE CASE (UNITY SHADER SCRIPT)

- Then let's talk about the 2 texture mix equations, check the **shader** of the 2 images on the right side.
- Here are 2 texture properties in the shader, and a slider that controls the interpolation T.



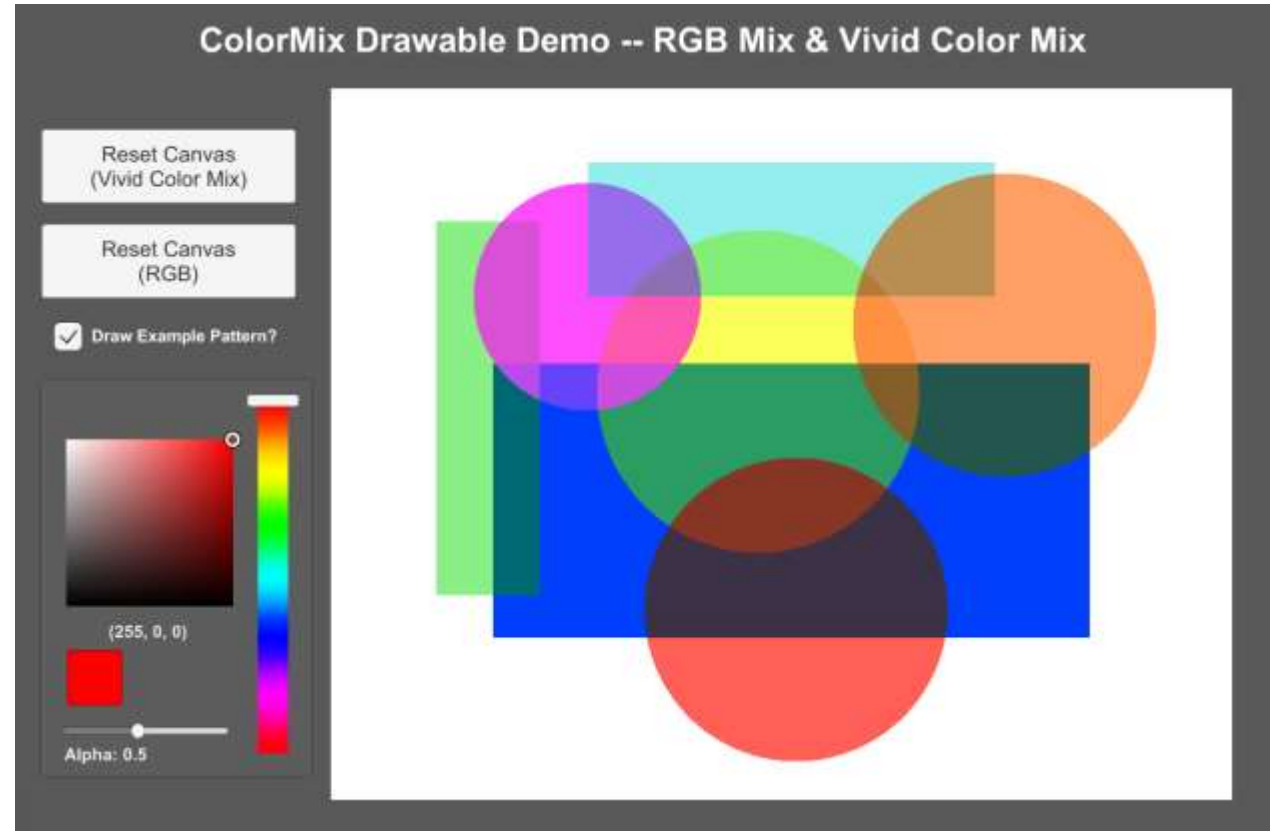
USE CASE (UNITY SHADER SCRIPT)

- You can **mix colors of the corresponding pixels** on the 2 textures by interpolation ratio T.
- We also provide **UI slider** to control the interpolation T.
- You can check it in script [ShaderSliderController](#).



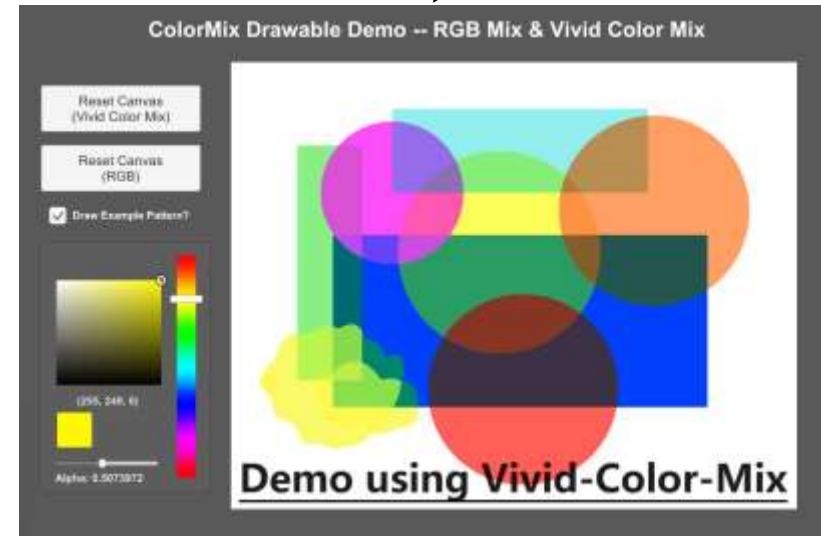
USE CASE (DRAWABLE DEMO)

- In the Scene **ColorMixDemoScene**, there is a drawable canvas, you can draw & try RGB-Mix and Vivid Color-Mix yourself.
- There is a **color picker** that you can select different colors and change color's alpha (used as concentration).
- You can change click/hold left mouse button to draw on canvas to check different colors' mix result in RGB-Mix or **Vivid Color-Mix**.



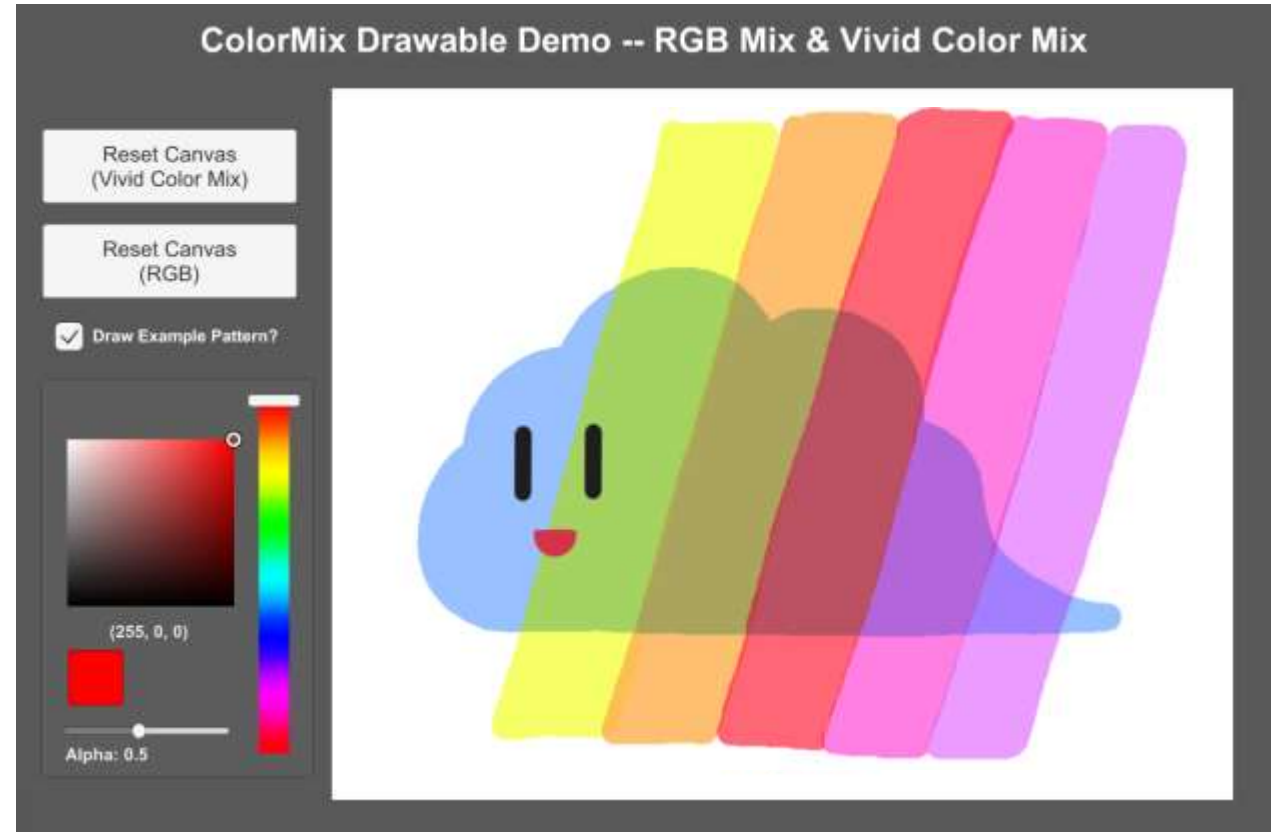
USE CASE (DRAWABLE DEMO)

- There are **2 buttons** to reset the canvas, respectively uses **Vivid Color-Mix** & **RGB-Mix**. You can click each button and check the differences between RGB-Mix and Vivid Color-Mix.
- There is also **a toggle** to control **whether to automatically draw sample patterns** when reset the canvas. If you want to check the mix differences, then select it. If you want to make your own paintings, then just cancel it.

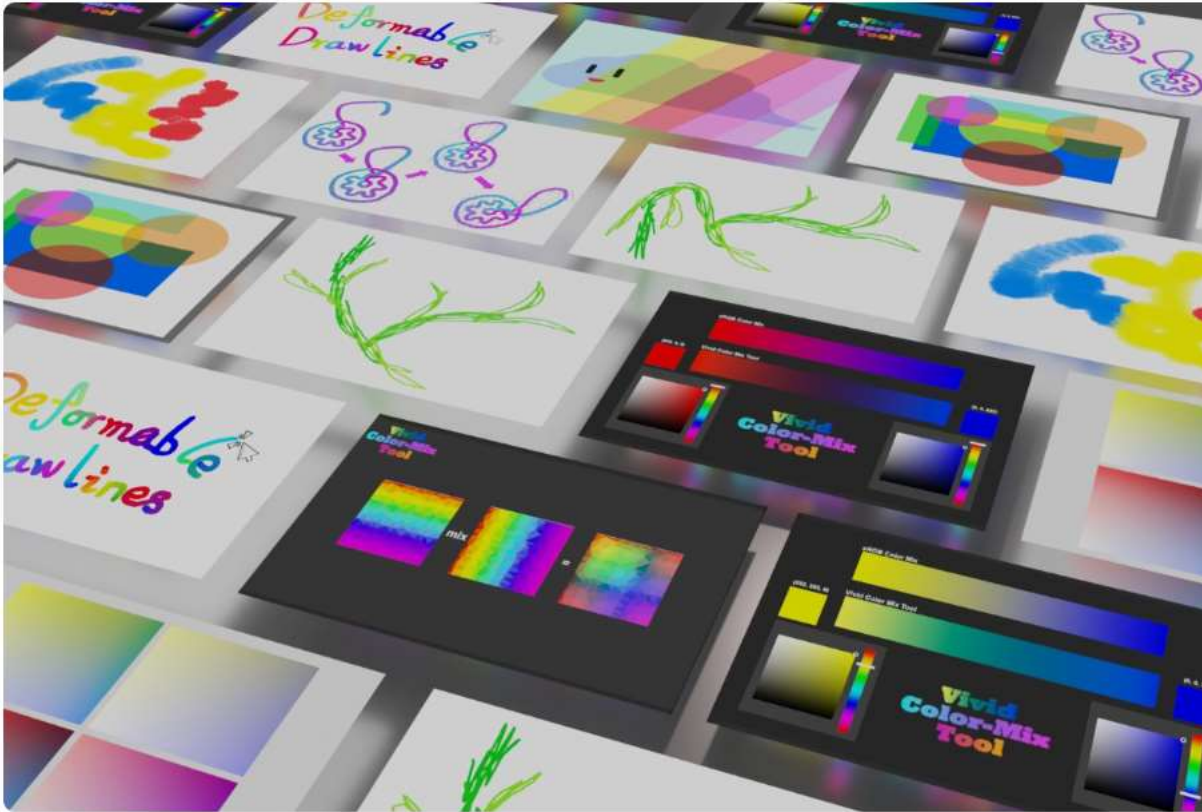


USE CASE (DRAWABLE DEMO)

- Try and make your own masterpiece!
- Use Vivid Color-Mix Method for a more excellent painting!



CREDITS



SpRiseMax-Chen

assetstore.unity.com/publishers/
website

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-Lightweight, Creative Painting Modules and Tools-

-- Research and development during my postgraduate period -- Including: 1. Deformable drawlines. 2. Color-mixing models. 3. Coming soon.

The programs are mainly from my postgraduate projects.

Rate me 5 stars and leave a like if it helps you, thanks!

SpRiseMaxChen 11.26.2023